A Leakage-Controlled Benchmark and Taxonomy-Aware Multi-Context Model for Fine-Grained Fish Species Classification on OzFish

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Abstract:

Reliable marine biodiversity monitoring relies, in part, on accurate identification of fish species in underwater surveys. Automated image analysis can reduce the tedium of manual annotation, but assessing progress requires certain evaluation protocols that account for the characteristics of survey data. Related observations from the same deployment, in particular, can compromise evaluation when distributed across training and testing sets, while overall accuracy may conceal poor recognition of underrepresented species and differences in the taxonomic severity of errors. To address these evaluation challenges, we introduce a curated fine-grained fish classification benchmark built on the existing OzFish dataset, which comprises of 76,083 fish crops across 321 species, 127 genera, and 39 families. The benchmark uses deployment-grouped training, validation, and testing partitions, and combines class-balanced accuracy with species-frequency and taxonomy-aware analyses. We evaluate eight classification systems over three random seeds using shared partitions and metrics, including a proposed Self-Distillation with No Labels version 3 (DINOv3)-based multi-context classifier with gated fusion and taxonomy-aware supervision. The proposed model achieves 69.0% macro accuracy and 84.2% overall accuracy, with mean predictive perfor-

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mance comparable to the evaluated Multi-context Attention and Taxonomy-Aware Network (MATANet) implementation with a much smaller parameter count. Ablation experiments done on the proposed model show that hierarchical supervision reduces taxonomic error severity while still maintaining similar species-level accuracy, whereas recognition of poorly represented species remains challenging across the evaluated systems. This benchmark establishes a common framework for comparing underwater fish classifiers and identifying limitations beyond aggregate accuracy.

Keywords: Fine-grained fish classification, Underwater imagery, Benchmark dataset, Long-tailed distribution, Taxonomy-aware learning, Data leakage
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1. Introduction

Marine biodiversity supports ecosystem functioning and the fisheries that we humans depend on. However, pressures such as overfishing, pollution, and habitat degradation threaten marine ecosystems, emphasizing the need for reliable monitoring systems to survey current fish species composition and abundance in order to guide preservationists and marine experts on further management decisions (Barbedo, 2022; Saleh et al., 2024). Underwater imagery provides a means of collecting these observations, but reviewing large volumes of footage requires considerable time and effort, as well as an expertise in taxonomy(Boom et al., 2014; Langlois et al., 2020). This creates a need for automated methods that can assist researchers in identifying fish species and processing survey imagery at scale (Barbedo, 2022; Li et al., 2022).

Deep learning models are increasingly developed and implemented to enable automated analysis of marine imagery, including fish detection and species classification(Saleh et al., 2022; Muksit et al., 2022; Guo et al., 2026). Their performance, however, depends in part on the characteristics and quality of the datasets used to train and evaluate them(Saleh et al., 2020; Khan et al., 2023). Underwater imagery presents a multitude of challenging imaging conditions, such as variable illumination, turbidity, and occlusion, which

can obscure the visual features needed to distinguish closely related species (Guo et al., 2026; Saleh et al., 2020; Thanjaivadivel and Thangaraj, 2026). Additionally, species are often unevenly represented within these datasets, producing long-tailed distributions in which a small number of classes account for most of the observations while many other species have relatively few examples (Khan et al., 2023; Van Horn et al., 2018; Felemban et al., 2026). This imbalance can limit recognition performance for poorly represented species, even when overall classification accuracy may appear high (Cui et al., 2019; Zhang et al., 2025). Evaluating performance across different species-frequency groups is necessary to understand how effectively a model can recognize well-represented and poorly represented classes (Felemban et al., 2026; Liu et al., 2019). The structure of the survey data also requires attention when defining evaluation protocols (Roberts et al., 2017; Beery et al., 2018). Multiple observations within the same deployment may contain closely related frames or stereo views (Langlois et al., 2020; Marrable et al., 2023), so assigning them independently to training and testing sets can produce overly optimistic estimates of performance on unseen deployments (Roberts et al., 2017; Beery et al., 2018; Kapoor and Narayanan, 2023).

These challenges highlight the necessity for benchmarks that evaluate recognition performance across unevenly represented species while also accounting for the structure of underwater survey data (Khan et al., 2023; Felemban et al., 2026; Kay et al., 2022). To support systematic evaluation under these conditions, we introduce a fine-grained fish classification benchmark built on the existing OzFish dataset (Australian Institute of Marine Science (AIMS) and University of Western Australia (UWA) and Curtin University, 2019). The curated benchmark contains 76,083 fish crops that span 321 species, 127 genera, and 39 families. We partition the data at the Baited Remote Underwater Video Station (BRUVS) deployment level (Langlois et al., 2020), keeping observations from the same deployment within a single partition, preventing overlap between training, validation, and testing sets (Roberts et al., 2017; Beery et al., 2018). The evaluation combines overall and class-balanced accuracy with species-frequency analyses (Liu et al., 2019) and taxonomy-aware metrics (Silla and Freitas, 2011; Bertinetto et al., 2020), providing complementary perspectives on model performance.

We evaluate eight different classification models using this benchmark, spanning convolutional networks, supervised vision transformers, a self-supervised representation baseline, and context-aware approaches. These include our proposed DINOv3-based model (Sim'eoni et al., 2025) that draws

inspiration from MATANet's multi-context architecture (Lee et al., 2026), combining target appearance with surrounding visual information through gated fusion. The model, hereafter referred to as Bio-TAC (Biological Taxonomy-Aware Classification), also incorporates taxonomy-aware supervision by deriving genus and family probabilities from its species predictions (Bertinetto et al., 2020; Lee et al., 2026). All models are evaluated using shared data partitions and a common set of classification and taxonomy-aware metrics, with results reported across three random seeds. Model parameter count was also compared between the different models alongside predictive performance. Additional ablation experiments on Bio-TAC investigate the effects of backbone adaptation, contextual inputs, hierarchical supervision, and long-tail training strategies.

2. Related Work

This section reviews prior work on underwater image datasets for fish recognition, fine-grained recognition and visual representations, environmental context in marine recognition, taxonomy-aware and long-tailed learning objectives, and evaluation protocols for ecological data. It closes by summarizing the gaps that motivate the benchmark and the model developed in this paper.

2.1. Underwater Image Datasets for Fish Recognition

Several public datasets have been developed for automated fish recognition, and they differ considerably in imaging conditions, label granularity, and intended task. Fish4Knowledge (Boom et al., 2014) was among the first large collections of in-situ underwater video, recorded by fixed cameras on coral reefs in Taiwan, and the LifeCLEF fish tasks (Joly et al., 2015) used this footage to organize fish identification as a shared evaluation challenge. WildFish (Zhuang et al., 2018) and its extension WildFish++ (Zhuang et al., 2021) are much broader taxonomically, reaching 2,348 categories, but both are built from web imagery whose appearance differs substantially from survey footage. DeepFish (Saleh et al., 2020) contains in-situ video from Australian habitats, although its classification task separates fish from background rather than identifying species. FathomNet (Katija et al., 2022) has become an important open resource for deep-sea imagery, providing taxonomically labelled detections across a wide range of midwater and benthic

taxa, and a curated subset of it was released for the FathomNet 2025 fine-grained recognition challenge at the CVPR 2025 FGVC12 workshop (Katija et al., 2022). FishNet (Khan et al., 2023) is larger still and pairs species labels with functional trait annotations. Broader overviews of underwater image datasets and fish recognition methods are given by Li et al. (2022) and Saleh et al. (2022) .

OzFish (Australian Institute of Marine Science (AIMS) and University of Western Australia (UWA) and Curtin University, 2019) is the dataset used in this work. It was produced under the Australian Research Data Commons Data Discoveries programme from more than three thousand historical BRUVS videos. Its full composition is reported in Section 3.1, and its position relative to the other resources reviewed here is summarised in Table 1. The annotations were made by fish ecologists following the marine sampling field manual for benthic stereo-BRUVS (Langlois et al., 2020), so the dataset combines survey-grade imagery with unusually deep and reliable taxonomic labels. So far, however, OzFish has been used mainly for detection and morphometry, for example in YOLO-Fish (Muksit et al., 2022) and in the semi-automated length measurement model of Marrable et al.(Marrable et al., 2023). No standard classification split has been released for the dataset, so studies that classify OzFish species adopt their own filtering thresholds and partitions, and their results are neither comparable nor reproducible. These ad hoc partitions also ignore the survey structure of the imagery, in which many crops originate from the same deployment, so the reported accuracies may be inflated by data leakage (Roberts et al., 2017; Kapoor and Narayanan, 2023). This work addresses both problems by building a curated, leakage-controlled fine-grained classification benchmark on OzFish and releasing it with a deployment-grouped split.

Table 1 compares the main public datasets for fish and marine species recognition with the benchmark proposed in this work. Counts are reported as provided by the original authors and may refer to images, cropped detections, or regions of interest (ROIs). The final column indicates whether each dataset is released with a split that keeps images from the same survey unit, site, or deployment within a single fold. As the final rows show, no earlier resource combines the taxonomic depth, survey-grade imagery, and released group-aware split that a rigorous fine-grained benchmark requires.

Table 1: Public datasets for fish and marine species recognition.

Dataset	Images / crops	Species	Taxonomic levels	Imagery source	Grouped split released
Fish4Knowledge (Boom et al., 2014)	27,370	23	Species	Fixed reef cameras, Taiwan	No
LifeCLEF2015-Fish (Joly et al., 2015)	9,160 ROIs	15	Species, order	Fixed reef cameras, Taiwan	No
WildFish (Zhuang et al., 2018)	54,459	1,000	Species	Web imagery	No
WildFish++ (Zhuang et al., 2020)	103,034	2,348	Species	Web imagery	No
DeepFish (Saleh et al., 2020)	~40,000	n/a	Fish vs. background	In-situ video, Australia	No
FathomNet2025 (Barnard et al., 2025)	23,700 ROIs	79 taxa	Mixed ranks	ROV and AUV, deep sea	No; test set withheld
FishNet (Khan et al., 2023)	94,532	17,357	Class to species	Aggregated databases	No; random 80/20
OzFish (AIMS et al., 2019)	~80,000 crops	507	Family, genus, species	Stereo-BRUVS, Australia	No
This work (OzFish benchmark)	76,083 crops	321	Family, genus, species	Stereo-BRUVS, Australia	Yes; deployment-grouped

2.2. Fine-Grained Recognition and Visual Representations

Fine-grained visual classification concerns categories that are visually similar to one another, as in the bird dataset CUB-200-2011 (Wah et al., 2011) and the multi-taxon iNaturalist dataset (Van Horn et al., 2018). In this setting the differences between classes are small and spatially localised, while variation within a class from pose, illumination, and background is large (Chang et al., 2021). Early deep learning approaches localised discriminative parts or modelled higher-order feature interactions (Lin et al., 2015; Fu et al., 2017; Zheng et al., 2017). Later work built on vision transformers, from the original ViT (Dosovitskiy et al., 2021) to TransFG (He et al., 2022), which selects discriminative tokens, and hierarchical variants such as Swin (Liu et al., 2021). Convolutional networks remain competitive, from

ResNet (He et al., 2016) through EfficientNetV2 (Tan and Le, 2021) to ConvNeXt (Liu et al., 2022) . Several of these architectures have been applied to fish recognition, including MSDBN (Si et al., 2024), SwinFishNet (Ergün, 2025), ConvFishNet (Qu et al., 2024), and the transformer study of Veiga and Rodrigues (2024). All of these models operate on the cropped fish alone, and each is evaluated on a differently constructed dataset, which makes their results hard to compare.

A parallel line of work has changed how the underlying representations are learned. DINO (Caron et al., 2021) showed that self-distillation without labels produces transformer features with strong object-centric structure. DINOv2 (Oquab et al., 2024) scaled this approach into general-purpose features that transfer well without fine-tuning, and DINOv3 (Sim'eoni et al., 2025) scaled it further, introducing Gram anchoring to keep dense patch features from degrading over long training schedules. The quality of these patch features matters for context modelling, because any method that attends over patch tokens depends on the tokens staying spatially sharp. Language-aligned models such as CLIP (Radford et al., 2021) and BioCLIP (Stevens et al., 2024), the latter trained on a taxonomically structured corpus, offer a complementary route. Freezing such a backbone and training a lightweight head on top of it is attractive in ecological applications, where labelled data are limited relative to the capacity of a large transformer (da Silva Martins et al., 2025; Kofidis et al., 2026).

Underwater imagery introduces degradations that these benchmarks do not contain, including wavelength-dependent color attenuation, turbidity and backscatter, marine snow, uneven illumination, motion blur, and frequent occlusion (Guo et al., 2026; Thanjaivadivel and Thangaraj, 2026; Jian et al., 2024). In BRUVS footage a fish often occupies only a small fraction of a high-resolution frame, so the resolution actually available for distinguishing species is far below the nominal image size (Langlois et al., 2020; Saleh et al., 2020). Despite these challenges, applied fish classification has been slow to adopt self-supervised backbones, and end-to-end fine-tuning of ImageNet-pretrained networks remains the norm (Saleh et al., 2022; Kofidis et al., 2026). How much adaptation such a backbone needs on BRUVS imagery is also unsettled. Frozen features and full fine-tuning are usually treated as alternatives, while the intermediate option of unfreezing only the last few transformer blocks is rarely reported, so the point at which further adaptation stops repaying its cost is unknown for this domain (Kofidis et al., 2026).

2.3. Environmental Context in Marine Recognition

Contextual information has long been known to support object recognition in computer vision(Torralba, 2003; Divvala et al., 2009; Zhang et al., 2020; Wang and Zhu, 2023). The effect is stronger in ecology, where habitat associations and co-occurrence patterns are part of how an expert identifies an ambiguous specimen. The same signal can also cause failures. Beery et al. (2018) showed on camera-trap data that models learn to rely on backgrounds and locations, and that accuracy drops sharply at sites not seen during training. Their later Context R-CNN (Beery et al., 2020) instead supplies context as an explicit input rather than letting the model absorb it implicitly. Within marine imagery, the closest work to the model proposed in this paper is MATANet (Lee et al., 2026). MATANet introduces a Multi-Context Environmental Attention Module (MCEAM), which applies cross-attention between the ROI representation and crops of the surrounding scene at 3Œ, 5Œ, and full-image scales, together with a Hierarchical Separation-Induced Learning Module (HSLM) that attaches an auxiliary classifier to each taxonomic level. In its ablation study on FathomNet2025(Katija et al., 2022), hierarchical distance improved from 2.24 with a single context stream to 1.90 with three streams, and to 1.77 once HSLM was added. Its strongest configuration reached 1.45 on the private challenge leaderboard, against 1.68 for the next-best published entry (Lee et al., 2026). Other entries to the same challenge also reported gains from combining full-image context with taxonomy-aware losses (Katija et al., 2022), which suggests the effect is not specific to one architecture.

The model evaluated in this paper adopts the MCEAM formulation and differs from MATANet (Lee et al., 2026) in three respects, described in Section 4. It is built on a substantially smaller backbone, hierarchical supervision is applied by marginalising a single species distribution instead of through per-level auxiliary classifiers, and evaluation is carried out on the proposed benchmark rather than on a curated challenge set. The last difference matters because existing context-aware results come from datasets whose partitions do not control for the survey unit an image came from, which leaves open how much of the reported gain is ecological evidence and how much is a site-identifying shortcut (Roberts et al., 2017; Beery et al., 2018; Salaudeen et al., 2025) . Capacity is a second confound. Where context is added to a backbone that is fine-tuned end to end, any improvement reflects both the contextual pathway and the extra trainable parameters introduced alongside it. The ablations in Section 6 therefore evaluate the contextual pathway un-

der a fixed frozen encoder, separating its effect from changes in backbone adaptation.

2.4. Taxonomy-Aware and Long-Tailed Learning Objectives

Biological taxonomy provides structure that a flat classifier ignores. Under standard cross-entropy, confusing two species of the same genus costs exactly as much as confusing two species from different families. Silla and Freitas (2011) divide methods that exploit this structure into local and global approaches. Local methods attach a classifier to each taxonomic level and train them jointly, as in the conditional softmax of YOLO9000(Redmon and Farhadi, 2017), the per-level design of MATANet (Lee et al., 2026), and FishAI (Yang et al., 2024) , which classifies marine fish at five taxonomic ranks and reports accuracy falling from 0.975 at class level to 0.626 at species level. Global methods impose the hierarchy through the loss or the label space while keeping a single output layer, as in Deng et al. (2012), the hierarchical cross-entropy of Bertinetto et al. (2020), Chang et al. (2021), and the hierarchical multi-label contrastive method of Zhang et al. (2022).

Local methods are simple to implement, but they do not model parent-child relations explicitly, and nothing forces the genus classifier to predict a genus that contains the species chosen by the species classifier(Silla and Freitas, 2011). Marginalising a single species distribution removes this inconsistency by construction, at the cost of placing the full distributional burden on the species head. This trade-off has not been examined closely against level-wise supervision in the marine setting. Evaluation has developed along similar lines, and lowest-common-ancestor metrics such as hierarchical distance now credit predictions that miss the species but recover the genus or family (Bertinetto et al., 2020; Katija et al., 2022; Deng et al., 2012).

Survey data are also long-tailed, exhibiting the skewed species-abundance distributions typical of ecological sampling, in which a few abundant species dominate the record while most are observed only rarely (Van Horn et al., 2018; Zhang et al., 2025). The usual response is reweighting, through focal loss (Lin et al., 2017), class-balanced weighting by effective sample number (Cui et al., 2019), or margin-based losses such as LDAM (Cao et al., 2019). Kang et al. (2020) showed that decoupling representation learning from classifier rebalancing often outperforms rebalancing both jointly, and Liu et al. (2019) addressed the open long-tailed setting directly. Resampling is the main alternative (Zhang et al., 2025), although class-balanced samplers

change both the exposure of each class and the number of optimisation steps per epoch, an interaction that is easy to overlook under a fixed epoch budget.

It is not yet clear which of these strategies transfer to marine data. On FishNet, class-balanced reweighting improved rare-class accuracy substantially, focal loss reduced it, and combining the two performed worse than class-balancing alone (Khan et al., 2023). This is a single result on a single dataset, but it indicates that long-tail methods validated on terrestrial benchmarks should be re-tested rather than assumed to work here. In applied fish classification the tail is more often removed than addressed (Saleh et al., 2022; Guo et al., 2026). Species below an occurrence threshold are discarded and results are reported as overall accuracy, which is dominated by the abundant classes, even though rare species are often those of greatest conservation interest (Santoro et al., 2025).

2.5. Evaluation Protocols and Data Leakage

The reliability of a benchmark depends on how its data are partitioned. Roberts et al. (2017) argue that when data carry spatial, temporal, hierarchical, or phylogenetic structure, random cross-validation seriously underestimates predictive error, and they recommend block cross-validation, in which the data are divided strategically rather than at random, wherever such dependence exists. Kapoor and Narayanan (2023) document how widespread the resulting leakage has become, identifying affected results in at least 294 papers across seventeen disciplines. Underwater survey video is a particularly difficult case. Frames from a single BRUVS deployment can show the same individual repeatedly, consecutive frames of the same animal, paired views from a stereo rig, and a substrate, lighting profile, and turbidity level that barely change (Langlois et al., 2020; Marrable et al., 2023). A partition drawn at the level of individual crops therefore places near-duplicates on both sides of the train and test boundary, and a model can score well simply by recognising the deployment rather than the species (Beery et al., 2018; Salaudeen et al., 2025). The concern is even greater for a model that is deliberately given wide-field context, because the background is exactly the signal that identifies a deployment (Beery et al., 2018; Salaudeen et al., 2025). Controlling for the survey unit is therefore a precondition for interpreting any claim that context improves recognition.

Grouped partitioning at the level of sites or sessions is established practice in the camera-trap literature (Beery et al., 2018) and follows directly from Roberts et al. (2017), and marine computer vision has started to adopt

it. The Caltech Fish Counting dataset (Kay et al., 2022) draws its training and test data from separate sonar camera deployments, so that generalisation is measured at unseen locations, and ReefNet (Felemban et al., 2026) pairs a conventional within-source split with a cross-source benchmark that withholds entire annotation sources. Fine-grained fish classification has not followed this practice. Random partitioning is still used even in flagship resources, with FishNet adopting a random 80/20 division of its images (Khan et al., 2023), and split definitions frequently remain unreleased, as the final column of Table 1 shows.

2.6. Summary of Research Gaps

The review above identifies four gaps that motivate this work. First, OzFish offers the taxonomic depth and survey provenance that a rigorous fine-grained benchmark requires, but no released, leakage-controlled classification split exists for it, so results reported on it are not comparable. Second, no study has compared convolutional networks, supervised transformers, self-supervised features under varying degrees of backbone adaptation, and a marine-specific context-aware model on the same underwater data under identical conditions. Third, the evidence that environmental context helps has come from datasets that do not control for the survey unit and from fully fine-tuned backbones, so the effect of context has not been separated from shortcut learning or from added model capacity. Fourth, taxonomy-aware supervision and long-tail treatment are usually studied separately, the prevailing hierarchical designs rely on per-level classifiers whose predictions need not agree across levels, and the long-tail methods most commonly applied have not been validated on marine imagery. This paper addresses these gaps by releasing a deployment-grouped OzFish classification benchmark, evaluating a panel of baselines on it under a common protocol, and proposing a taxonomy-aware multi-context classifier that enforces hierarchical consistency by marginalisation and achieves mean performance comparable to a much larger context-aware model at a fraction of its backbone size.

3. Dataset Specification

This section explains the construction of the OzFish classification benchmark used during this study. It details the source dataset, benchmark construction and filtering procedures, deployment-level splitting strategy, and resulting taxonomic and class distribution.

3.1. Dataset Source and Collection

OzFish is a publicly available dataset provided by the Australian Institute of Marine Science (AIMS)(Australian Institute of Marine Science (AIMS) and University of Western Australia (UWA) and Curtin University, 2019). The dataset contains around 80,000 fish crop images, 45,000 bounding box annotations, and capturing 507 species belonging to 200 genera and 70 families. Each sample in the dataset contains a hierarchical annotation, which contains which family, genus, and species the fish belongs to.

Given that the dataset was collected using Baited Remote Underwater Video Stations (BRUVS) (Langlois et al., 2020), makes the captured underwater imagery include challenging conditions, such as turbidity, motion blur, occlusions, cluttered backgrounds, etc (Saleh et al., 2022, 2020). Rather than laboratory conditions, since OzFish have these characteristics makes it a suitable benchmarking for fine-grained fish classification in realistic marine environments.

3.2. Dataset Curation

Several requirements need to be met in a dataset for it to be suitable for the benchmarking of fine-grained fish classification models. First, it must have a hierarchical taxonomic label (family, genus, and species) in order to help taxonomy-aware learning (Silla and Freitas, 2011; Bertinetto et al., 2020). Second, the dataset should be large enough to train transformer models, since transform-based models usually benefit from massive training datasets (Dosovitskiy et al., 2021; Veiga and Rodrigues, 2024). Finally, the dataset images should display realistic underwater conditions, such as motion blur, varying light brightness, occlusions, marine snow, and other environmental conditions, to make sure it is actually usable in real world scenarios (Guo et al., 2026; Saleh et al., 2020).

Ozfish dataset satisfies these requirements by including hierarchical taxonomic labels, a large number of annotated underwater images, and data captured using BRUVS under realistic underwater conditions. Therefore, it is well suited for benchmarking taxonomy-aware fine-grained classification and modern transformer-based architectures.

To create a standard and reproducible benchmark, the original OzFish dataset was curated by applying label cleaning and filtering species.

3.2.1. Label Cleaning

Prior to benchmark construction, the original OzFish metadata underwent a label cleaning process to ensure consistent and reliable taxonomic annotations. Since the benchmark aims to support taxonomy-aware fine-grained classification, inconsistencies in species naming or ambiguous labels could artificially increase the number of classes and negatively affect both model training and evaluation.

First, species labels were represented using their complete scientific binomial names, consisting of both the genus and species identifiers, rather than using the species epithet alone. This prevents different biological species from being merged into the same class when identical species epithets exist across different genera. In addition, known taxonomic spelling inconsistencies were standardized to a canonical representation to avoid duplicate class definitions caused by formatting or naming variations.

Furthermore, placeholder labels that do not correspond to valid species annotations were removed from the benchmark. Such labels provide ambiguous supervision and may introduce incorrect class information during training. By eliminating these samples and standardizing the remaining taxonomic annotations, the resulting benchmark provides a consistent and biologically meaningful label space suitable for taxonomy-aware learning.

3.2.2. Species Filtering

Even though the OzFish dataset contains a large number of unique species, 507 unique species to be exact, many of those species do not have enough samples for the model to learn about the species, since it cannot evaluate them fairly, having that the metrics would be unstable and unusable in this case, which leads to a poor train, validation, and testing representation. Therefore, inclusion rules had to be set in order to benchmark the OzFish dataset fairly and correctly. First, a species is included if it has at least 20 crops in the dataset, which is the minimum number of crops which provides sufficient samples for training and feasible dataset partitioning. Afterwards, the species should appear in at least three deployments, which will be explained in detail in Section 3.3.

Hence, Table 2 shows what are the final numbers of crops, species, genera, families, and deployments that were included in the final benchmark compared to the original OzFish.

Table 2: Comparison of the original OzFish dataset and the final benchmark.

Metric	Original Dataset	Final Benchmark
Crops	~80,000	76,083
Species	507	321
Genera	200	127
Families	70	39
Deployments	1,589	1,548

3.3. Dataset Splits

The chosen samples are partitioned based on their original BRUVS deployments in order to ensure that the benchmark offers reliable evaluation of model generalization. The subsections describe the motivation for deployment-level grouping, the stratification procedure, and leakage safe splitting.

3.3.1. Deployment-Level Splitting Strategy

An important part of creating a trustworthy image classification benchmark is making sure that the training, validation, and testing partitions remain statistically independent. Unlike conventional image datasets, since OzFish is collected using BRUVS (Langlois et al., 2020; Australian Institute of Marine Science (AIMS) and University of Western Australia (UWA) and Curtin University, 2019), performing a random crop level split allows highly similar images, or even the same individual fish, to appear at the same time in both the training and testing sets. Consequently, it introduces data leakage, which leads to overly positive performance estimates that do not accurately show the model's ability to generalize (Roberts et al., 2017; Beery et al., 2018; Kapoor and Narayanan, 2023).

Therefore, the proposed benchmark performs dataset partitioning at the single indivisible unit to eliminate this problem (Roberts et al., 2017). In order to ensure that every image originates from the same deployment is appointed only to either the training, validation, or testing partition, each BRUVS deployment is treated as a single indivisible unit. Using this method, it guarantees that stereo images pairs captured during the same deployment (Langlois et al., 2020; Marrable et al., 2023) remain within the same split, avoiding leakage between partitions.

3.3.2. Grouped Stratified Partitioning

While maintaining deployment integrity, the benchmark also seeks to preserve representative species distributions across the three partitions. Therefore, a grouped stratified splitting strategy was adopted, targeting an approximate 70%/15%/15% distribution for the training, validation, and testing sets, respectively. Since deployments contain varying numbers of images and species compositions, the final proportions cannot always match the target ratios exactly. Instead, the splitting algorithm prioritizes maintaining deployment integrity while distributing species as evenly as possible among the three partitions.

The resulting benchmark statistics are presented in Table 3, where the final dataset consists of 50,513 training crops, 12,458 validation crops, and 13,112 testing crops. These correspond to approximately 66.39%, 16.37%, and 17.23% of the complete benchmark, respectively. Despite the constraints imposed by grouped splitting, all 321 species are represented within the training partition, while the validation and testing partitions preserve the vast majority of the taxonomic diversity.

Table 3: Dataset Split Statistics

Split	Crops	Percentage	Species	Genera	Families	Deployments
Train	50,513	66.39%	321	127	39	1,054
Val	12,458	16.37%	313	125	38	264
Test	13,112	17.23%	311	123	39	230

The deployment statistics are summarized in Table 4. The benchmark contains 1,548 unique BRUVS deployments, with an average of 49.15 image crops per deployment and a median of 16 crops. The large variation in deployment sizes, ranging from a single crop to 630 crops, further demonstrates why deployment-level grouping is necessary, as random image-level partitioning would likely distribute highly correlated samples across different dataset splits.

Table 4: Deployment Statistics

Statistics	Value
Total Deployments	1548
Mean Crops per Deployment	49.15
Median Crops per Deployment	16
Min Crops per Deployment	1
Max Crops per Deployment	630

3.3.3. Leakage Verification

For the purpose of ensuring the integrity of the benchmark, following the partition generation an automated verification procedure was performed. The verification confirmed that no deployment appears in more than one dataset partition. In addition, Stereo image pairs are also guaranteed to remain within the same partition, since all samples originating from the same deployment remain grouped together. This leakage-safe splitting protocol provides a fair and reproducible evaluation benchmark that better reflects real-world model generalization than conventional random image-level splitting.

3.4. Taxonomic and Class Distribution

Following the dataset preparation, the benchmark includes 321 species belonging to 127 genera and 39 families, as shown in Table 5, which provides plenty of taxonomic hierarchy fitting for evaluating taxonomy aware classification models.

Table 5: Taxonomy Statistics

Metric	Value
Total Families	39
Total Genera	127
Total Species	321
Avg Species per Genus	2.53
Avg Genera per Family	3.26

As shown in Figure 1 and as many ecological datasets (Khan et al., 2023; Van Horn et al., 2018; Felemban et al., 2026), the benchmark shows noticeable long tail distribution. Therefore, in order to focus more on assessing the model's robustness across different representation frequencies, the benchmark reports classification performance separately for three frequency groups based on the number of training samples available for each species (Liu et al., 2019). In specific, species which contain more than 100 training samples are classed as Head, species between 21 and 100 training samples are classed into Medium, and species containing less than or equal to 20 training samples are classed as Tail (Liu et al., 2019). Each group's performance is reported independently, which allows researchers to learn whether the improvements are achieved equally across the benchmark or mainly on the well represented species.

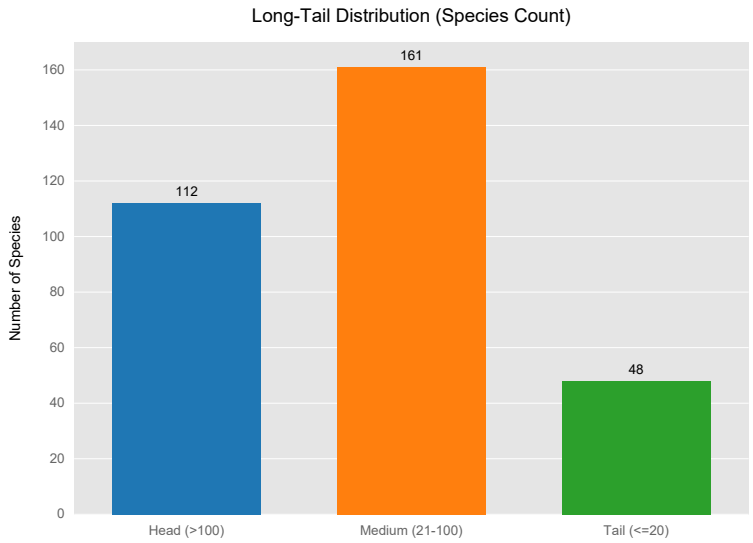


Figure 1: Long Tail distribution

As shown in Table 6, although 112 head species containing 82% of the total training samples, many of the species are represented by small number of observations, which results in an increase of difficulty of fine-grained recognition (Zhang et al., 2025).

Table 6: Long-Tail Analysis

Group	Species	Images	Percentage
Head (>100)	112	41,406	81.97%
Medium (21-100)	161	8,321	16.47%
Tail (<=20)	48	786	1.56%

4. Proposed Method

This paper proposes Bio-TAC, a taxonomy-aware multi-context classifier for fine-grained fish species recognition on the OzFish benchmark. The architecture in this section is presented using its baseline training configuration, while alternative training and adaptation strategies are evaluated experimentally in the later sections. Bio-TAC is built on a DINOv3 backbone (Sim'eoni

et al., 2025) that encodes four visual streams: the ROI view of the target fish, and the three progressively wider contextual views corresponding to the 3x, 5x, and full-frame scales (Lee et al., 2026).

The MCEAM layer (Lee et al., 2026) then fuses the target-fish representation with information from the three wider context views, producing a context-aware embedding. A learned gating mechanism then controls the relative contribution of context-aware and ROI-only representations before species classification.

4.1. Context-Stream Representation

For each target fish, four image streams are extracted from the complete frame using its bounding box $b = x,\ y,\ w,\ h$ (Lee et al., 2026). All local crops share the same bounding-box center. The ROI stream is the original bounding-box crop, representing the target fish's morphology. It may preserve discriminative features such as body shape, fin structure, and visible patterns, among other physical traits. The 3x context stream enlarges the bounding-box's width and height by a factor of three. While it contains the target fish, it also contains its immediate surroundings and may provide beneficial information about nearby fish and specific schooling patterns. The 5x context stream provides a wider view that may include habitat-associated contextual cues such as coral, sand, rocks, and reef structures. The full-frame stream provides global scene information, giving full context from the entire provided frame. The extracted views are processed using the common preprocessing and augmentation pipeline described in later sections.

4.2. DinoV3 Encoder

The four visual streams are encoded by a shared DINOv3 ViT-B/16 backbone (Sim'eoni et al., 2025; Dosovitskiy et al., 2021). Each 224x224 image is split into 16x16 patches, producing a 14x14 grid of 196 image patches. The backbone represents each token using a 768-dimensional embedding and produces CLS and patch-token representations for each stream. The CLS token is used as the target fish representation, while the patch tokens provide additional contextual features. The encoder weights are shared across all four streams, ensuring that the target and contextual features are represented within a common embedding space. The DINOv3 backbone is initialized from pretrained weights (Sim'eoni et al., 2025) and kept frozen in the baseline formulation, while the downstream fusion, projection, gating, and classification components are optimized on the OzFish training set.

4.3. Multi-Context Attention Fusion

The Multi-Context Environmental Attention Module (MCEAM) (Lee et al., 2026) fuses the ROI representation with features from the three wider context streams. The ROI CLS token is used as the query, as it represents the target fish, while the patch tokens from each context stream are used as keys and values, since they represent candidate contextual regions that the ROI representation may attend to, based on how useful the information is.

For context stream k, the attended representation is calculated as:

$$a^{(k)} = \text{softmax}\left(\frac{\mathbf{Q}\left(\mathbf{K}^{(k)}\right)^{\mathsf{T}}}{\sqrt{d_h}}\right)\mathbf{V}^{(k)} \tag{1}$$

where Q is obtained from the ROI CLS token, $\left(K^{(k)}\right)$ and $\left(V^{(k)}\right)$ are obtained from the patch tokens of context stream k, and d_h is the feature dimension of each attention head. The module uses eight attention heads and produces one attended representation for each of the three context streams.

The three attended context representations are concatenated with the original ROI representation. Since each representation contains 768 features, the concatenated vector has 3,072 dimensions. A feedforward fusion network then projects this vector into a 256-dimensional context-aware embedding, denoted by $z_{context}$.

To preserve a direct target-only representation, the ROI CLS token is also independently projected into a 256-dimensional ROI embedding, denoted by (z_{ROI}). A learned scalar gate determines the relative contribution of the context-aware and ROI-only representations for each sample. The final fused representation is calculated as:

$$\mathbf{z} = g\mathbf{z}_{context} + (1-g)\mathbf{z}_{ROI} \tag{2}$$

where g is produced by a sigmoid function and has a value between zero and one. A larger value of g gives greater weight to the context-aware representation, while a smaller value places greater emphasis on the target fish itself. The resulting 256-dimensional representation is subsequently passed to the species classification head.

4.4. Species Prediction and Taxonomic Marginalization

The final fused representation is passed through a linear classifier to produce logits over the S species classes:

$$\boldsymbol{\ell} = \mathbf{W}_c \mathbf{z} + \mathbf{b}_c \tag{3}$$

The logits are then converted into species probabilities using softmax:

$$p(s_i) = \frac{\exp(\ell_i)}{\sum_{j=1}^{S} \exp(\ell_j)} \tag{4}$$

The proposed model uses a single species-level prediction head. The genus and family probabilities get derived from the species distribution using known taxonomic relationships. For a certain genus g, the probabilities of all species that belong to that genus are summed:

$$p(g) = \sum_{s:\text{genus}(s)=g} p(s) \tag{5}$$

Similarly, the probability of a family f is obtained by:

$$p(f) = \sum_{s:\text{family}(s)=f} p(s) \tag{6}$$

This marginalization process makes sure that the species, genus, and family level probabilities stay consistent taxonomy-wise (Bertinetto et al., 2020; Deng et al., 2012). All higher-level probabilities are derived from the same species distribution, unlike other methods that use separate prediction heads for each taxonomic level (Silla and Freitas, 2011; Lee et al., 2026; Yang et al., 2024).

4.5. Taxonomy-Aware Training Objective

The OzFish training set has a long-tailed distribution, with large differences in the number of samples for each species. The species-level objective uses class-balanced (CB) focal loss (Cui et al., 2019; Lin et al., 2017) to reduce the dominance of very frequent classes.

For species c with n_c training samples, an effective-number weight is first calculated as(Cui et al., 2019):

$$\alpha_c = \frac{1-\beta}{1-\beta^{n_c}}. \tag{7}$$

The values are normalized across all of the species to obtain a final class weight w_c. Rare species therefore receive relatively greater weight than frequently observed species.

For a sample with true species y, the species-level loss is defined as (Cui et al., 2019; Lin et al., 2017):

$$\mathcal{L}_{species} = -w_y \left(1 - p(s_y)\right)^{\gamma} \log p(s_y), \tag{8}$$

where $p(s_y)$ is the probability that's assigned to the correct species and γ reduces the contribution of examples that have already been classified with high confidence. The values for β and γ are set as 0.9999 and 2 respectively (Cui et al., 2019; Lin et al., 2017).

The marginalized genus and family distributions provide additional supervision:

$$\mathcal{L}_{genus} = -\log p(g_y), \tag{9}$$

$$\mathcal{L}_{family} = -\log p(f_y), \tag{10}$$

where g_y and f_y denote the true genus and family that's associated with species y.

The complete training objective is:

$$\mathcal{L} = \lambda_s \mathcal{L}_{species} + \lambda_g \mathcal{L}_{genus} + \lambda_f \mathcal{L}_{family}, \tag{11}$$

with reference weights set to:
$\lambda_s = 1,\ \lambda_g = 2,\ \lambda_f = 3.$

The genus and family level terms encourage the predicted probability mass to stay within the correct higher-level taxonomic groups, while the CB focal term addresses the great imbalance between common and rare species that is found in OzFish.

5. Benchmark Protocol and Experimental Setup

This section describes the benchmark comparison protocol, training and implementation settings, evaluation metrics, and ablation design used in the experiments. The experimental setup was designed to provide consistent comparisons across the evaluated models while accommodating method-specific requirements wherever it was deemed necessary.

5.1. Benchmark Models and Comparison Protocol

For this benchmark paper, we evaluated 8 different models representing different architectural families on the OzFish benchmark for comparison. The conventional CNN baselines selected for this benchmark include ResNet-50 (He et al., 2016), ConvNeXt-Tiny (Liu et al., 2022), and EfficientNetV2-S (Tan and Le, 2021), while the transformer-based models are ViT-B/16 (Dosovitskiy et al., 2021) and Swin-B (Liu et al., 2021). We also included DINOv3 ViT-B/16 linear probe, a self-supervised representation baseline (Sim'eoni et al., 2025), as well as MATANet, which provides a specialized fine-grained and context-aware comparison (Lee et al., 2026). Finally, Bio-TAC which is our proposed taxonomy-aware multi-context approach.

The comparison protocol for the different architectures follows a specific ruleset necessary for the benchmark. There were some common aspects across the evaluation of all models to maintain consistency across the experiments, such as the train/val/test split which was kept consistent, as well as the metrics that were used to evaluate each model's performance. The experimental conditions were standardized wherever possible, and every model solves the same species classification task. There were, however, some method-specific differences that remained due to each model's differing design and architecture.

There is an important distinction between the standard baselines and Bio-TAC, however, as all of the single crop baselines rely solely on the ROI crop, meaning that it primarily focuses on the target fish morphology, while Bio-TAC has access to the target morphology as well as the surrounding visual information taken from the 3 extra visual crops, therefore feeding more information to Bio-TAC in comparison to the other single-crop baselines. This distinction is important when interpreting the benchmark results, as we are comparing complete classification systems against each other instead of isolating the effect of any single architectural component. The contribution of the extra context streams is separately examined through the ablation experiments that are described in Section 5.4.

One exception to the rest of the models is MATANet (Lee et al., 2026). Since MATANet's official implementation was released by its authors, it was used directly rather than being reimplemented. We used the same OzFish benchmark split and converted the samples into the format that MATANet's code expects, then we trained MATANet using its intended architecture and native training procedure to then evaluate the produced species predictions using the common benchmark evaluation metrics. MATANet was not forced

to follow the common training configuration since there are method-specific details that may not be common with the other model implementations.

5.2. Training and Implementation Details

Unless otherwise specified, the following preprocessing and training settings apply to models implemented within the common benchmark framework. MATANet retained its native implementation and training procedure as described separately below (Lee et al., 2026). Before the images were passed to the models, they had to be prepared using preprocessing and augmentation methods. The images were all converted into a standardized 224x224 input resolution. aspect-ratio-preserving letterbox resizing was used to preserve the morphology of the target fish without distorting its aspect ratio, with bicubic interpolation used during resizing and zero padding added wherever necessary to obtain the required square input. The RGB images that resulted from this preprocessing were then normalized using the standard ImageNet mean and standard deviation. data augmentation was also introduced during training to increase the variation in fish orientation and underwater image appearance. Geometric augmentation consisted of horizontal flipping and rotations between -30 degrees and +30 degrees, while vertical flipping was excluded, as it would produce orientations that are uncommon in the original imagery. We included additional appearance-based perturbations like synthetic marine-snow artifacts, motion blur, Gaussian noise, and some brightness, contrast, and saturation changes. For Bio-TAC, the same geometric transformation was applied to all the four visual streams to preserve spatial consistency. There was no stochastic augmentation applied during validation or testing.

The models implemented within the common benchmark framework were trained with a 60-epoch budget using the AdamW optimizer. A fixed batch size of 32 was used, as well as a weight decay of 0.01 and a learning rate of $1x10^{-4}$, with the exception of Bio-TAC's learning rate being $1x10^{-5}$ to provide more conservative updates during the full fine-tuning of the pretrained DINOv3 backbone. A three-epoch warm-up period was used rather than applying the full learning rate from the beginning, during which the learning rate was increased gradually to its configured value. After the warm-up period, the learning rate gets progressively reduced using a cosine annealing schedule for the remaining portion of the training period. To reduce GPU memory requirements and improve computational efficiency, we also implemented automatic mixed-precision training.

The pretrained backbones had differing treatment across the evaluated models. Regarding the DINOv3 linear-probe baseline (Sim'eoni et al., 2025), its backbone remained frozen throughout the training period, and only the species-classification layer was optimized. Contrastingly, the pretrained ResNet-50 (He et al., 2016), EfficientNetV2-S (Tan and Le, 2021), ConvNeXt-Tiny (Liu et al., 2022), ViT-B/16 (Dosovitskiy et al., 2021), and Swin-B (Liu et al., 2021) baselines were fine-tuned end-to-end on the OzFish training set. As for Bio-TAC, the pretrained DINOv3 backbone (Sim'eoni et al., 2025) was fine-tuned jointly with the MCEAM module (Lee et al., 2026), projection layers, gating mechanism, and species-classification head, which allows for both the visual representation as well as the newly introduced components to adapt to the OzFish classification task.

We maintained an exponential moving average (EMA) of the trainable model parameters throughout training using a decay coefficient of 0.999, which was also used during validation. At the end of each epoch, the model was evaluated on the validation partition, and the checkpoint that had achieved the lowest validation Mean Hierarchical Distance (MHD) (Bertinetto et al., 2020; Katija et al., 2022) was retained as the best-performing checkpoint. MHD was used consistently as the predefined checkpoint-selection criterion, while the complete set of classification and taxonomy-aware metrics was reported on the test partition. We did not apply early stopping, meaning that all of the models completed the full 60-epoch training schedule regardless of whether or not the score started declining after it reached the best validation performance. The retained checkpoint was used for final evaluation on the test partition after the training had completed.

To account for the stochastic nature of model training, each configuration was trained independently using three random seeds: 0, 1, 2. The train/val/test partitions remained fixed across all of the runs, which meant that the different seeds only affected the stochastic training operations, such as parameter initialization, data ordering, and data augmentation, allowing for the assessment of the stability of each model without introducing variation from different dataset partitions. The experimental configurations were stored in YAML files, with the software environment and source-code revision recorded, supporting reproducibility.

MATANet's implementation has its own specific training procedure, and it was treated separately from the models trained within the common benchmark framework (Lee et al., 2026). The model was trained using a DINOv2-

Large backbone (Oquab et al., 2024) with an input resolution of 256x256. The model had an effective batch size of 16, a learning rate of $1x10^{-6}$, and a weight decay of 0.001 over a 60-epoch training schedule. The native data transformations as well as the hierarchical training procedure for MATANet were retained, while the same OzFish benchmark partition and common evaluation metrics were used, thus ensuring that the final predictions it generates stayed comparable with the other evaluation methods.

5.3. Evaluation Metrics and Statistical Reporting

Due to the strong long-tailed class distribution (Zhang et al., 2025; Liu et al., 2019) and hierarchical taxonomy (Silla and Freitas, 2011; Bertinetto et al., 2020) of the OzFish benchmark, we evaluated model performance using several complementary metrics. Conventional classification metrics were used as a way to quantify species-level recognition performance. Taxonomy-aware metrics were also used to assess the severity within the taxonomic hierarchy of any incorrect prediction. Performance was examined separately across head, medium, and tail species to determine how well each evaluated model dealt with varying levels of training-data availability. All benchmark results are reported as the mean ± standard deviation across the three independent runs described in Section 5.2.

5.3.1. Classification Performance

To determine the performance of classification for each evaluated model, three different accuracy metrics were used: Macro Accuracy, top-1 accuracy, and top-5 accuracy. Since OzFish has a long-tailed class distribution, overall accuracy metrics may favour frequent species, therefore Macro Accuracy was used as the primary class-balanced species-level performance metric reported in the benchmark in order to give equal importance to each of the species by first calculating the accuracy separately for each species and then averaging the species accuracies (Liu et al., 2019; Kang et al., 2020). The formula for Macro Accuracy is as follows:

$$\text{MacroAcc} = \frac{1}{C} \sum_{c=1}^{C} \frac{TP_c}{N_c}, \tag{12}$$

Where C represents the number of evaluated species, TP_c represents the number of correctly classified samples belonging to species c, and N_c represents the total test samples belonging to species c.

Top-1 accuracy was also used as a classification metric to measure if the highest scoring species out of all predictions made by the evaluated models is correct. Top-1 is sample-weighted rather than class-balanced, as every sample has equal influence.

Top-5 accuracy was utilized to measure if the ground-truth species was among the five highest scoring species predictions that the evaluated model produced. Since fine-grained classification involves visually similar species, the model may assign high probability to several plausible closely related classes, which is why top-5 accuracy proves a useful complementary metric in the case that the model assigned the ground-truth species one of its five highest scores, even if it wasn't ranked first.

5.3.2. Taxonomy-Aware Performance

Taxonomic error severity was evaluated using a hierarchical error score based on the family-genus-structure of the OzFish benchmark (Bertinetto et al., 2020; Katija et al., 2022; Deng et al., 2012). For a ground-truth species y and predicted species $\widehat{y}$, the hierarchical distance is defined as:

$$d(y,\widehat{y}) = \begin{cases} 0, & \widehat{y} = y, \\ 1, & \widehat{y} \neq y, \text{ same genus}, \\ 2, & \text{different genera within the same family}, \\ 3, & \text{the species belong to different families.} \end{cases} \tag{13}$$

MHD was then calculated by averaging this score over all test samples:

$$\text{MHD} = \frac{1}{N}\sum_{i=1}^{N} d(y_i, \widehat{y}_i), \tag{14}$$

Where N is the number of test samples, y_i is the ground-truth species for sample i, and $\widehat{y}_i$ is the corresponding predicted species. A lower MHD value indicates better overall taxonomic performance, resulting from a combination of more correct predictions and errors that remain closer to the ground-truth species within the taxonomic hierarchy.

Another metric that was used was Mean Mistake Severity (MMS), as it was used to isolate the severity of incorrect predictions (Bertinetto et al., 2020). Unlike MHD, correctly classified samples are excluded from this calculation, meaning that the metric only measures the average taxonomic distance of the mistakes made by the model.

Genus and family accuracy were additionally reported by checking whether the top-1 predicted species belonged to the same genus or family as the ground-truth species.

5.3.3. Long-Tail Performance

As the OzFish benchmark follows a long-tailed class distribution, it was important to also evaluate model performance according to the number of training samples available for each species. For benchmark-specific long-tail analysis, species were divided into three frequency groups based on their training-set sample counts, with head species containing more than 100 training samples, medium species containing between 21 and 100 samples, and tail species containing 20 or fewer samples (Liu et al., 2019).

Performance within each group was calculated using Macro Accuracy computed separately for the species belonging to that group, where the accuracy of each species was independently calculated before averaging across the species that belonged to the corresponding frequency group. This prevents more frequently represented species within a certain group from having a greater influence on the reported value. The resulting accuracies of the head, medium, and tail groups provide a more detailed view of how each evaluated model performs under different levels of training-data availability, particularly for rare species where limited training data makes recognition more challenging (Zhang et al., 2025; Liu et al., 2019).

5.4. Ablation Protocol

A series of matched ablation experiments was conducted to further examine the principal architecture and training choices of Bio-TAC. The ablations were organized into controlled comparison groups rather than treating every experiment as a direct modification of a single reference configuration. Within each comparison group, all experimental settings were fixed except for the factor that was under investigation.

The contribution of the multi-context pathway was examined under a fixed frozen-DINOv3 configuration. The complete three-context MCEAM configuration (Lee et al., 2026) was compared with an ROI-only variant, where both the contextual streams and the MCEAM were removed. A variant with reduced-context that retained only the 3x contextual stream was also compared with the complete 3x, 5x, and full-frame configuration as a way to examine the contribution of the multi-scale contextual information. Under

the same configuration, the depth of MCEAM varied between one, two, and four attention blocks.

The backbone adaptation was evaluated separately at a fixed learning rate of $1x10^{-4}$(Kofidis et al., 2026) The frozen DINOv3 encoder was compared with configurations in which the final two transformer blocks, the final four blocks, or the complete backbone were trainable. A separate learning-rate comparison was subsequently performed for the fully fine-tuned backbone using $1x10^{-4}$ and $1x10^{-5}$.

The effects of taxonomy-aware supervision and species-level loss formulation were evaluated under the fully fine-tuned DINOv3 configuration using a matched 2x2 objective design. The species-level objective used either CB focal loss (Cui et al., 2019; Lin et al., 2017) or conventional cross entropy, while hierarchical supervision (Bertinetto et al., 2020; Lee et al., 2026) was independently enabled or disabled, allowing the contribution of hierarchical supervision and class-imbalance-aware loss to be examined separately while still retaining the same architecture and training configuration.

Finally, a separate matched backbone comparison replaced DINOv3 (Sim'eoni et al., 2025) with DINOv2 (Oquab et al., 2024) while retaining the remaining full-fine-tuning configuration. Since the corresponding DINOv2 and DINOv3 encoders are not architecturally identical, this experiment is interpreted as a comparison of pretrained backbones rather than a strict isolation of the pretraining method.

6. Results and Discussion

The analysis and presentation of the experimental results obtained on the proposed OzFish benchmark are presented in this section. Starting the analysis with the comparison between the main classification models, followed by frequency-group performance, comparison with MATANet, ablation studies, qualitative analysis, computational efficiency, and finally a summary of the overall analysis.

6.1. Main Benchmark Results

This section presents the results of all the tested methods on the proposed OzFish benchmark. Results are reported as mean ± standard deviation over three independent seeds. The evaluation includes Macro Accuracy, Top-1, MHD, MMS, and cumulative Genus- and Family-level Accuracy, as shown in Table 7.

Table 7: Performance comparison of the evaluated models on the OzFish benchmark. The best score in each column is shown in bold, and the second-best score is underlined.

Model	Macro Accuracy	Top-1	MHD	MMS
Linear Probe	0.471±0.002	0.675±0.001	0.764±0.003	2.349±0.001
ResNet-50	0.571±0.007	0.749±0.002	0.539±0.005	2.145±0.006
EfficientNetV2-S	0.624±0.013	0.782±0.005	0.447±0.010	2.054±0.006
ConvNeXt-T	0.662±0.002	0.809±0.003	0.373±0.004	1.948±0.016
ViT-B	0.658±0.010	0.808±0.005	0.377±0.011	1.966±0.006
Swin-B	0.689±0.009	0.820±0.003	0.347±0.008	1.931±0.019
MATANet	**0.692±0.012**	**0.844±0.002**	**0.288±0.008**	**1.843±0.059**
Proposed Model	0.690±0.023	0.842±0.006	0.291±0.008	1.844±0.018

Model	Genus Accuracy	Family Accuracy
Linear Probe	0.754±0.001	0.807±0.001
ResNet-50	0.833±0.002	0.879±0.002
EfficientNetV2-S	0.866±0.002	0.905±0.002
ConvNeXt-T	0.890±0.001	0.928±0.001
ViT-B	0.888±0.004	0.927±0.002
Swin-B	0.900±0.002	0.933±0.003
MATANet	**0.917±0.006**	0.951±0.003
Proposed Model	0.916±0.002	**0.951±0.001**

As the backbone of conventional image classification baselines become more capable, they show a consistent improvement. Where ResNet-50 (He et al., 2016) achieves a Macro Accuracy of 0.571, while EfficientNetV2-S (Tan and Le, 2021) and ConvNeXt-T (Liu et al., 2022) increase this to 0.624 and 0.662, respectively. A similar performance is seen with transformer-based baselines, with ViT-B achieving 0.658 (Dosovitskiy et al., 2021) and Swin-B (Liu et al., 2021) reaching 0.689 Macro Accuracy. The trend shows that the benchmark remains difficult enough for performance to be affected by differences in visual representation capabilities.

The proposed model achieves a macro accuracy of 0.690 and a Top-1 accuracy of 0.842, making it competitive with the strongest benchmark baselines. More importantly, it achieves a MHD of 0.291 and MMS of 1.844, together with Genus and Family accuracies of 0.916 and 0.951, respectively. These results show that, in addition to producing accurate species prediction, the proposed method tends to make taxonomical closer mistakes when its prediction is incorrect.

The results also show that Top-1 accuracy alone is not enough to describe

the model quality, because the benchmark is fine-grained and taxonomically structured. Since models with similar Top-1 accuracy score can have huge difference in the severity of their classification errors, therefore, encouraging the use of hierarchical evaluation metrics in addition to traditional accuracy metrics (Bertinetto et al., 2020).

6.2. Performance Across Frequency Groups

Table 8: Macro Accuracy across head, medium, and tail species groups, defined by training-set frequency. The best score in each column is shown in bold, and the second-best score is underlined.

model	Head	Medium	Tail
Linear Probe	0.629±0.003	0.430±0.003	0.260±0.005
ResNet-50	0.730±0.005	0.517±0.001	0.397±0.030
EfficientNetV2-S	0.779±0.002	0.577±0.009	0.441±0.060
ConvNeXt-T	0.804±0.003	0.628±0.009	0.467±0.014
ViT-B	0.803±0.002	0.625±0.014	0.460±0.015
Swin-B	0.817±0.001	0.656±0.012	**0.521±0.018**
MATANet	<u>0.821±0.004</u>	**0.681±0.017**	<u>0.466±0.017</u>
Proposed Model	**0.843±0.004**	<u>0.668±0.008</u>	0.444±0.110

To further examine the effect of the long-tailed species distribution, Table 8 reports the macro accuracy across frequency groups Head, Medium, and Tail according to their training-set frequency (Liu et al., 2019). A more detailed understanding of whether the overall benchmark performance can be a representative of both the common and rare species is provided by this analysis.

As the species frequency decreases, a performance decline is noted across all models, indicating that classifying rare species is still substantially more difficult than classification of frequently observed species (Khan et al., 2023; Zhang et al., 2025). The proposed model performs well on the head and medium frequency species with a macro accuracy of 0.843 and 0.668, respectively, while its tail performance remains unstable with a macro accuracy of 0.444. This shows that even with the improvements attained by the proposed architecture, challenges connected to limited training samples are still not completely eliminated. The frequency group results showcase a crucial exchange in the proposed model. It achieved the highest head group frequency performance compared to the remaining models and keeps close performance in medium group frequency species. The tail group result, on

the other side, is lower and shows significantly greater variance among seeds. This indicates that while extremely rare species continue to be a significant benchmark limitation, the suggested architecture is effective for the dominating and moderately represented classes.

6.3. Comparison with MATANet

MATANet (Lee et al., 2026), introduced in Section 2.3, is the closest context- and taxonomy-aware comparator. Therefore, the comparison is considered useful for determining whether the proposed approach brings competitive performance under the same benchmark conditions.

The proposed model and MATANet show very similar performance across the reported evaluation metrics. The proposed model obtains a Macro Accuracy of 0.690 compared with 0.692 for MATANet, while Top-1 Accuracy is 0.842 compared with 0.844. Their MHD are similarly close, at 0.291 for the proposed model and 0.288 for MATANet. Genus- and family-level accuracies are also nearly identical, reaching 0.916 and 0.951 for the proposed model, compared with 0.917 and 0.951 for MATANet.

These comparable results are obtained despite differences in model scale. MATANet uses a fine-tuned DINOv2 large backbone (Oquab et al., 2024), on the other hand, the proposed model employs a smaller DINOv3 based backbone (Sim'eoni et al., 2025) together with multi-context attention and hierarchical supervision. Therefore, the proposed model reached a similar performance of classification and taxonomic consistency while running with a substantially smaller backbone.

In conclusion, the results indicate that the proposed model offers a competitive alternative to MATANet on the benchmark, with the main difference not coming from a significant difference in overall accuracy but rather from the balance between predictive performance and model scale.

6.4. Ablation Studies

The ablation studies look into the specific aspects and design choices that make up the proposed model. The internal experimental configurations utilized to separate the impacts of backbone adaptation, contextual modeling, hierarchical supervision, and long-tail methods are included in this section, in contrast to the primary benchmark comparison.

6.4.1. Backbone Adaptation

The impact of adapting the task-specific backbone is first assessed by increasing the number of trainable blocks inside the pretrained DINOv3 backbone (Sim'eoni et al., 2025). As shown in Figure 2, just from unfreezing the last two blocks, the Macro Accuracy increases from 0.507 ± 0.006 to 0.618 ± 0.006 and reduces MHD from 0.632 ± 0.012 to 0.421 ± 0.004. Further increasing the fine-tuning to the last four blocks increases Macro Accuracy to 0.657 ± 0.008 and reduces MHD to 0.385 ± 0.008.

Finally, fine-tuning the full backbone further increases Macro Accuracy to 0.677 ± 0.004 and reduces Mean Hierarchical Distance to 0.368 ± 0.003. The results indicate that increasing the number of backbone adaptations significantly improves taxonomic error quality and species classification (Kofidis et al., 2026).

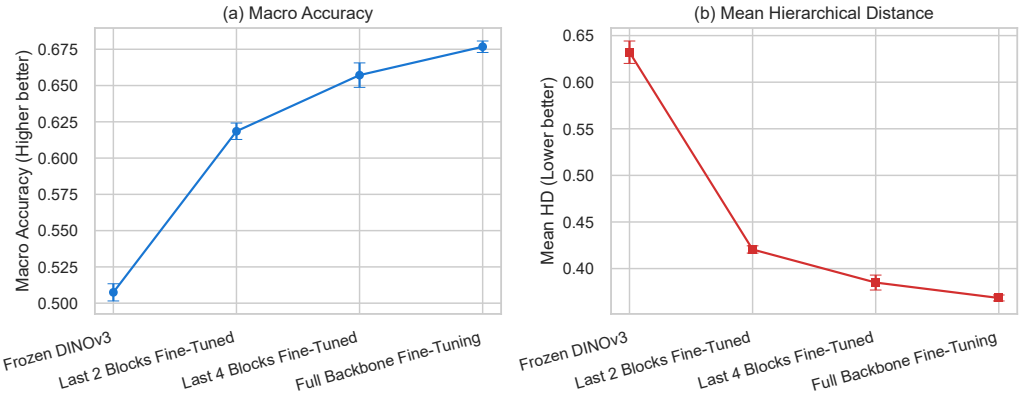


Figure 2: Impact on fine-grained OzFish classification of progressive DINOv3 backbone adaptation. Results are reported as mean ± standard deviation over three seeds for a frozen backbone, fine-tuning of the last two blocks, fine-tuning of the last four blocks, and full backbone fine-tuning. (a) Macro Accuracy, (b) Mean Hierarchical Distance, where more taxonomically consistent errors are indicated by lower values.

6.4.2. Taxonomy-Aware Objective

The effect of hierarchical supervision is evaluated by comparing flat species-level classification with the taxonomy-aware objective under the same unfrozen training settings. As shown in Figure 3, both produce similar species level performance, with a Top-1 score of 0.844 ± 0.005 for the flat objective and 0.842 ± 0.006 for the taxonomy-aware objective.

On the other hand, the difference is more shown in hierarchical measures. The taxonomy-aware objective reduces MHD from 0.314 ± 0.011 to 0.291 ± 0.008 and MMS from 2.009 ± 0.025 to 1.844 ± 0.018. Family Accuracy increases from 0.939 ± 0.003 to 0.951 ± 0.001, while Genus Accuracy increases from 0.903 ± 0.004 to 0.916 ± 0.002.

The results show that rather than significant improvement in exact species-level accuracy, the taxonomy-aware objective's main contribution is improved taxonomic consistency (Bertinetto et al., 2020). Therefore, while preserving comparable species-level classification performance, integrating the species, genus, and family relationship into the objective increases the quality of incorrect predictions.

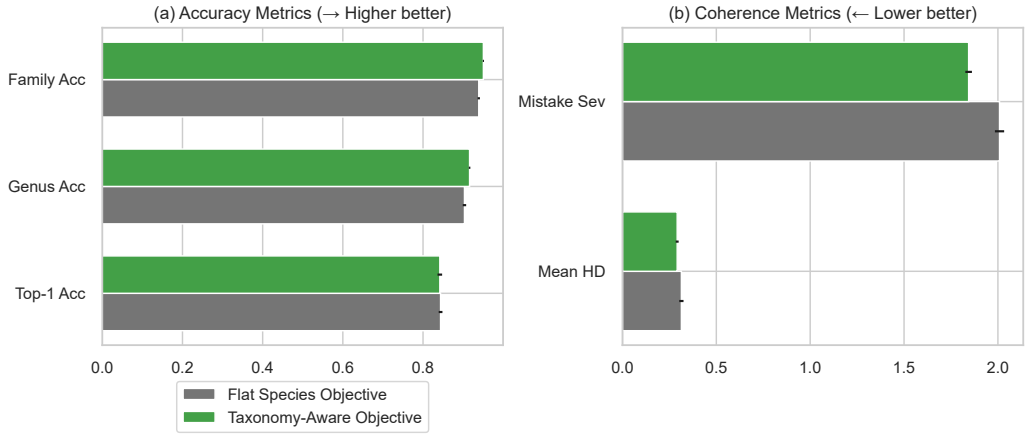


Figure 3: Comparison of flat species-level and taxonomy-aware objectives. (a) Reports species and taxonomy level accuracy metrics, (b) reports MMS and MHD.

6.4.3. Context Modeling

By comparing the target fish representation with increasingly broader spatial context, the influence of contextual visual information is tested. The ROI only configuration includes tight fish crops without any MCEAM based contextual streams, whereas the broader configuration uses social and habitat information around the target fish.

Using the same frozen DINOv3 setting, as shown in Figure 4(a), ROI only configuration achieves a Macro Accuracy of 0.466 ± 0.006, which increases to 0.512 ± 0.002 when social context is introduced. While the full-context configuration achieves a Macro Accuracy of 0.507 ± 0.006. The results show

that integrating contextual information does in fact provide significant improvement rather than using the fish ROI only, however, adding multiple contextual streams does not necessarily produce an increase in Macro Accuracy.

Another thing that should be assessed is the effect of adding more MCEAM attention depth, which is also tested using the same frozen DINOv3 settings and by Macro Accuracy metric. As shown in Figure 4(b), increasing the depth from one to two attention blocks increases Macro Accuracy from 0.507 ± 0.006 to 0.512 ± 0.008, while increasing the depth to four blocks reduces it to 0.505 ± 0.004. This implies that increasing the depth does not correspond to improved classification performance.

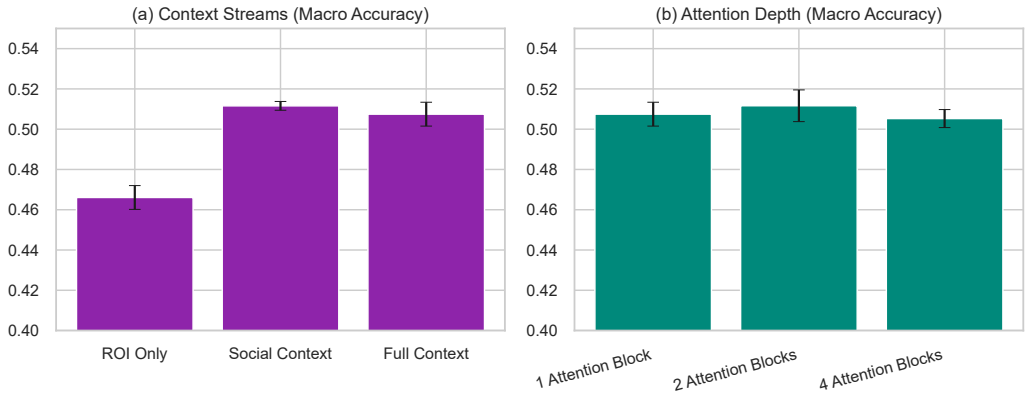


Figure 4: Ablation of contextual representation and MCEAM depth under the frozen DINOv3 configuration. (a) Compares the different context configurations, from only ROI classification to broader contextual inputs. (b) Effect of increasing MCEAM attention depth from one to four blocks. Results are reported as mean ± standard deviation over three seeds.

6.4.4. Long-Tail Components

In order to address the severe class imbalance in the OzFish dataset, the evaluation across the head, medium, and tail frequency groups under the frozen DINOv3 setting was needed to show the impact of different sampling strategies and loss functions.

As shown in Figure 5(a), using balanced sampling rather than random sampling slightly improves the performance on rare species, increasing Macro Accuracy on the tail classes from 0.316 ± 0.044 to 0.328 ± 0.036, and on the

medium classes from 0.465 ± 0.005 to 0.481 ± 0.004. On the other hand, this exchange comes with a noticeable cost to the dominant species, reducing Macro Accuracy on the head classes from 0.660 ± 0.011 to 0.638 ± 0.009.

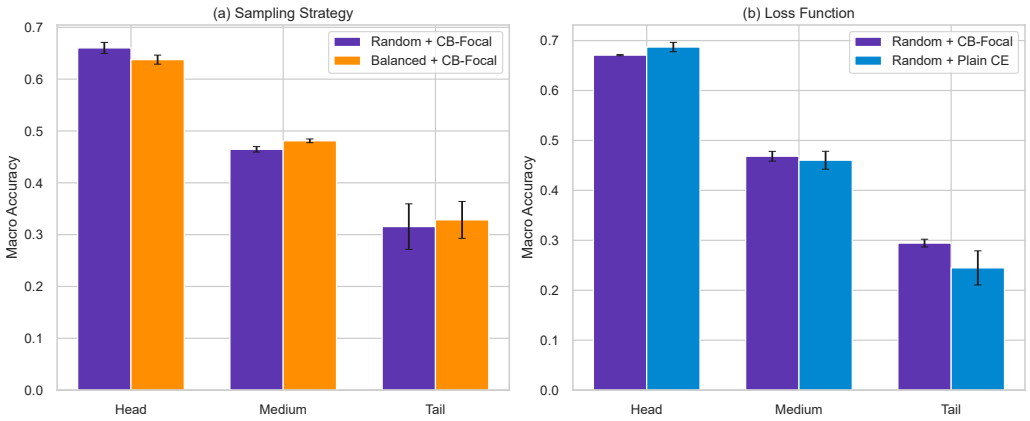


Figure 5: Effect of long-tail training strategies on species-frequency groups under the same frozen DINOv3 configuration. Results are reported as mean ± standard deviation over three seeds. (a) Comparison between random and balanced sampling strategies. (b) Effect of applying Class-Balanced Focal Loss (CB-Focal) compared to Plain Cross-Entropy (Plain CE).

Figure 5(b) shows the effect of using different loss functions, comparing Plain Cross-Entropy (Plain CE) against Class-Balanced Focal Loss (CB-Focal) (Cui et al., 2019; Lin et al., 2017) . As shown, CB-Focal loss gives significant increase in the rare species, increasing the tail Macro Accuracy from 0.245 ± 0.034 to 0.294 ± 0.008. Although the head class performance is slightly decreased, from 0.687 ± 0.009 to 0.671 ± 0.001, the significant gain on the long tail is highly advantageous. The results indicate that even though balanced sampling failed to achieve enough improvement to validate its cost on frequent classes, integrating CB-Focal loss when the backbone capacity is restricted could lessen the dataset's long tail issue.

6.5. Qualitative Analysis

By examining both the representative prediction of the Proposed Model and the attention pattern obtained from its multi-context architecture, this qualitative analysis was made. Figure 6 reports two representative test samples: a correctly classified *Lethrinus punctulatus* example and a misclassified

Caranx ignobilis example predicted as Gnathanodon speciosus. Where the figure also shows the MCEAM cross-attention maps related to social, habitat, and full-frame context streams, as well as backbone attention rollout for the ROI for each sample.

The misclassified sample illustrates that contextual information does not necessarily result in accurate species classification under visually challenging conditions. In addition, the attention patterns include the surrounding environment regions along side the target fish, indicating the different visual information which the multi-context representation includes. On the other hand, the correctly classified demonstrates how the model's ability to adapt to various spatial regions surrounding the target fish by using the model's contextual streams.

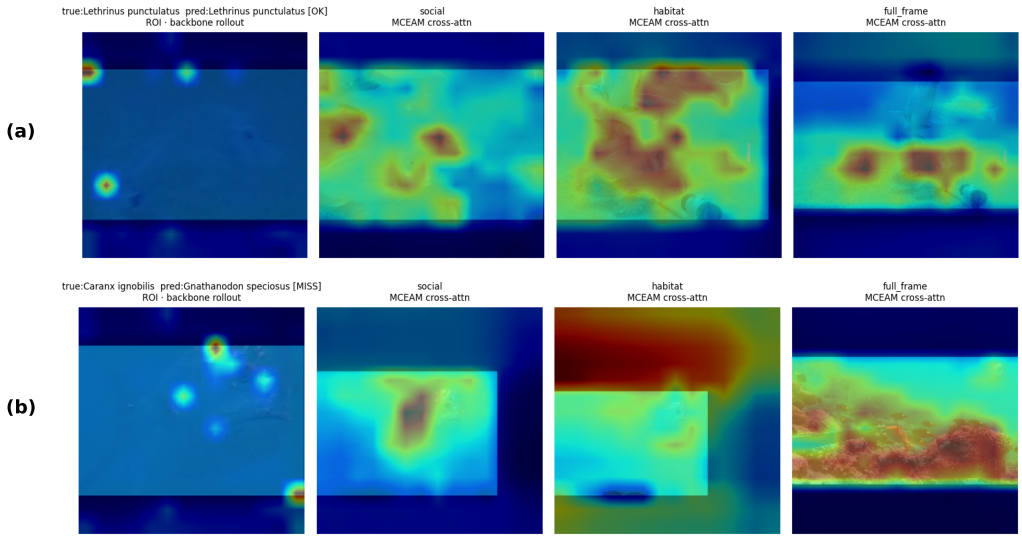


Figure 6: Qualitative predictions and multi-context attention visualizations from the Proposed Model. (a) Correct classification of Lethrinus punctulatus. (b) Misclassification of Caranx ignobilis as Gnathanodon speciosus.

These visualization shows how the Proposed Model distributes attention across the target in addition to its surroundings. They are intended to complement the quantitative context and classification analyses rather than serve as causal explanations of the model's predictions.

6.6. Computational Efficiency

Table 9: Model size and predictive performance on the OzFish benchmark: total parameter count, Macro Accuracy, and Mean Hierarchical Distance (MHD).

Model	Total Parameters	Macro Accuracy	MHD
Linear Probe	85,907,265	0.471 ± 0.002	0.764 ± 0.003
ResNet-50	24,165,761	0.571 ± 0.007	0.539 ± 0.005
EfficientNetV2-S	20,588,689	0.624 ± 0.013	0.447 ± 0.010
ConvNeXt-T	28,066,977	0.662 ± 0.002	0.373 ± 0.004
ViT-B	86,045,505	0.658 ± 0.010	0.377 ± 0.011
Swin-B	87,072,249	0.689 ± 0.009	0.347 ± 0.008
MATANet	1,347,468,072	0.692 ± 0.012	0.288 ± 0.008
Proposed Model	98,157,378	0.690 ± 0.023	0.291 ± 0.008

Table 9 reports the computational performance of the evaluated models. MATANet (Lee et al., 2026) contains 1,347,468,072 parameters, where on the other hand, the Proposed Model contains 98,157,378 parameters, equivalent to approximately one-fourteenth of the total number of parameters. Even though the huge difference in model size, their benchmarking performance is still almost the same, with MATANet achieving a Macro Accuracy of 0.692 ± 0.012 and a MHD of 0.288 ± 0.008, compared with 0.690 ± 0.023 and 0.291 ± 0.008, respectively, for the Proposed Model.

Another model that is worth referencing is Swin-B (Liu et al., 2021), which provides a parameter count similar to the Proposed Model, equal to 87,072,249 parameters. It achieves a Macro Accuracy of 0.689 ± 0.009 and a MHD of 0.347 ± 0.008, showing similar overall classification accuracy but a higher hierarchical error than the Proposed Model.

The results show that the Proposed Model was able to perform comparable to MATANet while using approximately 7.3% of its total parameters, emphasizing a huge different balance between classification performance and model scale.

6.7. Overall Findings

The decisions made in order to reach the final Proposed model came from the benchmark and ablation experiments. Among the architectural changes that were assessed, the one with the largest improvement was the backbone adaptation, progressively greater fine-tuning of the DINOv3 backbone

resulted in an increase in Macro Accuracy and decrease in MHD. Additionally, the ablation studies indicated that contextual information also provides significant improvements over ROI-only classification, on the other hand, increasing the attention depth over the initial settings did not result in constant benefits.

A different type of improvement was observed in the taxonomy-aware objective ablation. Even though Top-1 species level accuracy remained almost similar to the flat objective, however, integrating the taxonomy hierarchy decreased both metrics, MHD and MMS, while increasing genus and family level accuracy. Since hierarchical supervision's primary contribution was to improve taxonomic consistency rather than simply an increase in exact species accuracy (Bertinetto et al., 2020), this influenced the decision to include it in the final model.

The ablation that further affected the selection of the final strategy was long-tail experiments. Using the frozen DINOv3 as the main setting, CB-focal (Cui et al., 2019; Lin et al., 2017) resulted in better performance in the tail-group than plain cross-entropy while both use random sampling, showing that when the backbone remains restricted, using explicit class-imbalance handling is beneficial. However, the final Proposed Model uses a fully fine-tunned DINOv3 backbone, therefore, using plain cross entropy and random sampling achieved significantly better tail results than the frozen configuration. The results suggest that the long-tail strategy performance depends on the representation-learning approach (Khan et al., 2023; Kang et al., 2020). Under constrained frozen backbone training, CB-focal displays noticeable improvements, on the other hand, using the full adapted model perform better without requiring an additional loss modification.

As a result, the final model configuration combined the parts that consistently contributed to the ablation study: task-specific backbone adaptation, multi-context representation, taxonomy-aware supervision, and a simpler plain cross-entropy objective. The Proposed Model achieves 0.690 ± 0.023 Macro Accuracy and 0.291 ± 0.008 MHD on the benchmark, while keeping performance equivalent to MATANet at a significantly smaller model scale.

7. Limitations

Due to the nature of this work, there were certain limitations that we faced during the benchmark, specifically within the curated Ozfish bench-

mark itself. The main limitation that had to be considered was the fact that the dataset has a long-tail distribution, and even though there were measures taken to mitigate class imbalance, the results still showed that the models often underperformed with tail species due to their small sample sizes. The deployment-aware split also introduces limitations for the rarest species. Since some species are represented by a small number of observations or deployments, they cannot always be represented across all dataset partitions, which may reduce the stability of validation and test estimates for low-frequency classes, although maintaining deployment independence is important to avoid overly optimistic performance caused by highly related samples appearing in multiple folds (Roberts et al., 2017; Kapoor and Narayanan, 2023).

Another important limitation to discuss is the geographic specificity of the benchmark. All the experiments that the models were evaluated under were based on the species and ecology of the Australian marine ecosystems represented in the curated OzFish benchmark, meaning that they may not generalize well in a different region, as each region differs in its ecological distribution as well as environmental conditions (Felemban et al., 2026; Beery et al., 2018). Evaluation on independent datasets from other marine regions would be necessary to assess model generalization across different environments beyond OzFish.

The taxonomy-aware evaluation also provides a simplified representation of error severity. Hierarchical errors are categorized according to species, genus, and family membership, allowing taxonomically closer and more distant mistakes to be distinguished consistently. This ordinal measure, however, does not represent continuous phylogenetic or ecological distance between species.

Lastly, the ablation analysis also has a limitation in that several architectural ablations, including the context and MCEAM-depth comparisons, were conducted under a frozen DINOv3 configuration, whereas the final proposed model uses full backbone fine-tuning. These experiments therefore isolate the corresponding architectural choices within their matched experimental setting, but do not fully characterize possible interactions between contextual fusion and backbone adaptation.

8. Conclusion and future work

This paper introduced a fine-grained species classification benchmark built on the OzFish dataset, together with a taxonomy-aware multi-context classifier evaluated on it. OzFish combines survey-grade BRUVS imagery with family, genus, and species annotations, but it has been released without a standard classification split, so published results on it are neither directly comparable nor protected against leakage between the highly correlated frames that a single deployment produces. The benchmark addresses this by curating the dataset to 76,083 crops spanning 321 species, and by partitioning it at the level of the BRUVS deployment rather than the individual crop, so that stereo pairs and near-duplicate frames of the same individual cannot fall on both sides of the train and test boundary. The resulting split is released so that later work can be evaluated on identical data. The proposed model encodes the target fish alongside three progressively wider views of its surroundings, fuses them through cross-attention under a learned gate that weighs contextual evidence against target morphology, and obtains genus- and family-level probabilities by marginalising a single species distribution, which keeps predictions consistent across taxonomic levels by construction.

Eight models spanning convolutional, transformer, self-supervised, and context-aware families were evaluated over three seeds under a common protocol, using a metric panel that combines conventional accuracy with taxonomy-aware and frequency-stratified measures. Performance rose steadily with backbone capability across the conventional baselines, from a macro accuracy of 0.571 for ResNet-50 to 0.689 for Swin-B, indicating that the benchmark remains difficult enough to separate representations of differing strength. The proposed model reached a macro accuracy of 0.690 and a Top-1 accuracy of 0.842, with an MHD of 0.291 and a family-level accuracy of 0.951, achieving mean performance comparable to MATANet across the headline metrics while using a considerably smaller backbone. The metric panel also proved necessary rather than decorative, since models with comparable Top-1 accuracy differed measurably in the taxonomic severity of their errors. Accuracy fell consistently as species frequency decreased, and tail performance was consistently the weakest and generally more variable across seeds, which remains the principal limitation of the benchmark alongside its restriction to Australian marine ecosystems.

Three directions follow from these results. The clearest remaining headroom lies in the tail, where reweighting alone proved insufficient and where

methods that adjust the classifier rather than the loss, such as logit adjustment or decoupled classifier retraining (Kang et al., 2020), are worth testing on this benchmark. A second direction is temporal, as BRUVS footage provides short clips around each annotated frame, and aggregating predictions over a sequence rather than a single view may recover species that one degraded frame cannot resolve (da Silva Martins et al., 2025). Finally, applying the benchmark protocol to survey data from other marine regions (Felemban et al., 2026) would establish whether the deployment-grouped design and the observed ranking of methods hold beyond Australian waters. By releasing the split, the protocol, and the baseline results, this work aims to give subsequent studies of fine-grained marine recognition a common and leakage-controlled basis for comparison.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT to assist with manuscript organization, language and presentation, LaTeX preparation, reference and consistency checks, and review against journal submission requirements. The authors reviewed and verified the resulting assistance, edited the manuscript as needed, and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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Data and code availability

The original OzFish dataset used in this study is publicly available from the Australian Institute of Marine Science at `https://doi.org/10.25845/5e28f062c5097`. The curated benchmark splits, model implementations, experiment configurations, training and evaluation code, and reported results are publicly available at `https://github.com/Karam-Kreidli/ozfish-finegrained-classification-benchmark`.

CRediT authorship contribution statement

Karam Berhan Kreidli: Conceptualization, Methodology, Software, Validation, Investigation, Data curation. **Alaa Alrasheed:** Conceptualization, Formal analysis, Writing – original draft, Writing – review & editing, Visualization. **Mohamed Ayham:** Conceptualization, Writing – original draft, Writing – review & editing. **Kheireddine Sebihi:** Conceptualization, Writing – original draft, Writing – review & editing. **Manar Abu Talib:** Resources, Supervision, Project administration, Funding acquisition, Writing – review & editing. **Sohail Abbas:** Supervision, Project administration. **Ayad Turky:** Supervision, Project administration. **Manar Anwer Abusirdaneh:** Supervision, Writing – review & editing.

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